Seeing Like a State

How Certain Schemes to Improve the Human Condition Have Failed

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Introduction

This book grew out of an intellectual detour that became so gripping that I decided to abandon my original itinerary altogether. After I had made what appeared to be an ill-considered turn, the surprising new scenery and the sense that I was headed for a more satisfying destination persuaded me to change my plans. The new itinerary, I think, has a logic of its own. It might even have been a more elegant trip had I possessed the wit to conceive of it at the outset. What does seem clear to me is that the detour, although along roads that were bumpier and more circuitous than I had foreseen, has led to a more substantial place. It goes without saying that the reader might have found a more experienced guide, but the itinerary is so peculiarly off the beaten track that, if you're headed this way, you have to settle for whatever local tracker you can find.

A word about the road not taken. Originally, I set out to understand why the state has always seemed to be the enemy of "people who move around," to put it crudely. In the context of Southeast Asia, this promised to be a fruitful way of addressing the perennial tensions between mobile, slash-and-burn hill peoples on one hand and wet-rice, valley kingdoms on the other. The question, however, transcended regional geography. Nomads and pastoralists (such as Berbers and Bedouins), hunter-gatherers, Gypsies, vagrants, homeless people, itinerants, runaway slaves, and serfs have always been a thorn in the side of states. Efforts to permanently settle these mobile peoples (sedentarization) seemed to be a perennial state project—perennial, in part, because it so seldom succeeded.

The more I examined these efforts at sedentarization, the more I came to see them as a state's attempt to make a society legible, to arrange the population in ways that simplified the classic state functions of taxation, conscription, and prevention of rebellion. Having begun to think in these terms, I began to see legibility as a central problem in statecraft. The premodern state was, in many crucial respects, partially blind; it knew precious little about its subjects, their wealth, their landholdings and yields, their location, their very identity. It lacked anything like a detailed "map" of its terrain and its people. It lacked, for the most part, a measure, a metric, that would allow it to "translate" what it knew into a common standard necessary for a synoptic view. As a result, its interventions were often crude and self-defeating.

It is at this point that the detour began. How did the state gradually get a handle on its subjects and their environment? Suddenly, processes as disparate as the creation of permanent last names, the standardization of weights and measures, the establishment of cadastral surveys and population registers, the invention of freehold tenure, the standardization of language and legal discourse, the design of cities, and the organization of transportation seemed comprehensible as attempts at legibility and simplification. In each case, officials took exceptionally complex, illegible, and local social practices, such as land tenure customs or naming customs, and created a standard grid whereby it could be centrally recorded and monitored.

The organization of the natural world was no exception. Agriculture is, after all, a radical reorganization and simplification of flora to suit man's goals. Whatever their other purposes, the designs of scientific forestry and agriculture and the layouts of plantations, collective farms, ujamaa villages, and strategic hamlets all seemed calculated to make the terrain, its products, and its workforce more legible —and hence manipulable—from above and from the center.

A homely analogy from beekeeping may be helpful here. In premodern times the gathering of honey was a difficult affair. Even if bees were housed in straw hives, harvesting the honey usually meant driving off the bees and often destroying the colony. The arrangement of brood chambers and honey cells followed complex patterns that varied from hive to hive—patterns that did not allow for neat extractions. The modern beehive, in contrast, is designed to solve the beekeeper's problem. With a device called a "queen excluder," it separates the brood chambers below from the honey supplies above, preventing the queen from laying eggs above a certain level. Furthermore, the wax cells are arranged neatly in vertical frames, nine or ten to a box, which enable the easy extraction of honey, wax, and propolis. Extraction is made

possible by observing "bee space"—the precise distance between the frames that the bees will leave open as passages rather than bridging the frames by building intervening honeycomb. From the beekeeper's point of view, the modern hive is an orderly, "legible" hive allowing the beekeeper to inspect the condition of the colony and the queen, judge its honey production (by weight), enlarge or contract the size of the hive by standard units, move it to a new location, and, above all, extract just enough honey (in temperate climates) to ensure that the colony will overwinter successfully.

I do not wish to push the analogy further than it will go, but much of early modern European statecraft seemed similarly devoted to rationalizing and standardizing what was a social hieroglyph into a legible and administratively more convenient format. The social simplifications thus introduced not only permitted a more finely tuned system of taxation and conscription but also greatly enhanced state capacity. They made possible quite discriminating interventions of every kind, such as public-health measures, political surveillance, and relief for the poor.

These state simplifications, the basic givens of modern statecraft, were, I began to realize, rather like abridged maps. They did not successfully represent the actual activity of the society they depicted, nor were they intended to; they represented only that slice of it that interested the official observer. They were, moreover, not just maps. Rather, they were maps that, when allied with state power, would enable much of the reality they depicted to be remade. Thus a state cadastral map created to designate taxable property-holders does not merely describe a system of land tenure; it creates such a system through its ability to give its categories the force of law. Much of the first chapter is intended to convey how thoroughly society and the environment have been refashioned by state maps of legibility.

This view of early modern statecraft is not particularly original. Suitably modified, however, it can provide a distinctive optic through which a number of huge development fiascoes in poorer Third World nations and Eastern Europe can be usefully viewed.

But "fiasco" is too lighthearted a word for the disasters I have in mind. The Great Leap Forward in China, collectivization in Russia, and compulsory villagization in Tanzania, Mozambique, and Ethiopia are among the great human tragedies of the twentieth century, in terms of both lives lost and lives irretrievably disrupted. At a less dramatic but far more common level, the history of Third World development is littered with the debris of huge agricultural schemes and new cities (think of Brasília or Chandigarh) that have failed their residents.

It is not so difficult, alas, to understand why so many human lives have been destroyed by mobilized violence between ethnic groups, religious sects, or linguistic communities. But it is harder to grasp why so many well-intended schemes to improve the human condition have gone so tragically awry. I aim, in what follows, to provide a convincing account of the logic behind the failure of some of the great utopian social engineering schemes of the twentieth century.

I shall argue that the most tragic episodes of state-initiated social engineering originate in a pernicious combination of four elements. All four are necessary for a full-fledged disaster. The first element is the administrative ordering of nature and society—the transformative state simplifications described above. By themselves, they are the unremarkable tools of modern statecraft; they are as vital to the maintenance of our welfare and freedom as they are to the designs of a would-be modern despot. They undergird the concept of citizenship and the provision of social welfare just as they might undergird a policy of rounding up undesirable minorities.

The second element is what I call a high-modernist ideology. It is best conceived as a strong, one might even say muscle-bound, version of the self-confidence about scientific and technical progress, the expansion of production, the growing satisfaction of human needs, the mastery of nature (including human nature), and, above all, the rational design of social order commensurate with the scientific understanding of natural laws. It originated, of course, in the West, as a by-product of unprecedented progress in science and industry.

High modernism must not be confused with scientific practice. It was fundamentally, as the term "ideology" implies, a faith that borrowed, as it were, the legitimacy of science and technology. It was, accordingly, uncritical, unskeptical, and thus unscientifically optimistic about the possibilities for the comprehensive planning of human settlement and production. The carriers of high modernism tended to see rational order in remarkably visual aesthetic terms. For them, an efficient, rationally organized city, village, or farm was a city that *looked* regimented and orderly in a geometrical sense. The carriers of high modernism, once their plans miscarried or were thwarted, tended to retreat to what I call miniaturization: the creation of a more easily controlled micro-order in model cities, model villages, and model farms.

High modernism was about "interests" as well as faith. Its carriers, even when they were capitalist entrepreneurs, required state action to realize their plans. In most cases, they were powerful officials and heads of state. They tended to prefer certain forms of planning and so-

cial organization (such as huge dams, centralized communication and transportation hubs, large factories and farms, and grid cities), because these forms fit snugly into a high-modernist view and also answered their political interests as state officials. There was, to put it mildly, an elective affinity between high modernism and the interests of many state officials.

Like any ideology, high modernism had a particular temporal and social context. The feats of national economic mobilization of the belligerents (especially Germany) in World War I seem to mark its high tide. Not surprisingly, its most fertile social soil was to be found among planners, engineers, architects, scientists, and technicians whose skills and status it celebrated as the designers of the new order. High-modernist faith was no respecter of traditional political boundaries; it could be found across the political spectrum from left to right but particularly among those who wanted to use state power to bring about huge, utopian changes in people's work habits, living patterns, moral conduct, and worldview. Nor was this utopian vision dangerous in and of itself. Where it animated plans in liberal parliamentary societies and where the planners therefore had to negotiate with organized citizens, it could spur reform.

Only when these first two elements are joined to a third does the combination become potentially lethal. The third element is an authoritarian state that is willing and able to use the full weight of its coercive power to bring these high-modernist designs into being. The most fertile soil for this element has typically been times of war, revolution, depression, and struggle for national liberation. In such situations, emergency conditions foster the seizure of emergency powers and frequently delegitimize the previous regime. They also tend to give rise to elites who repudiate the past and who have revolutionary designs for their people.

A fourth element is closely linked to the third: a prostrate civil society that lacks the capacity to resist these plans. War, revolution, and economic collapse often radically weaken civil society as well as make the populace more receptive to a new dispensation. Late colonial rule, with its social engineering aspirations and ability to run roughshod over popular opposition, occasionally met this last condition.

In sum, the legibility of a society provides the capacity for large-scale social engineering, high-modernist ideology provides the desire, the authoritarian state provides the determination to act on that desire, and an incapacitated civil society provides the leveled social terrain on which to build.

I have not yet explained, the reader will have noted, why such high-

modernist plans, backed by authoritarian power, actually failed. Accounting for their failure is my second purpose here.

Designed or planned social order is necessarily schematic; it always ignores essential features of any real, functioning social order. This truth is best illustrated in a work-to-rule strike, which turns on the fact that any production process depends on a host of informal practices and improvisations that could never be codified. By merely following the rules meticulously, the workforce can virtually halt production. In the same fashion, the simplified rules animating plans for, say, a city, a village, or a collective farm were inadequate as a set of instructions for creating a functioning social order. The formal scheme was parasitic on informal processes that, alone, it could not create or maintain. To the degree that the formal scheme made no allowance for these processes or actually suppressed them, it failed both its intended beneficiaries and ultimately its designers as well.

Much of this book can be read as a case against the *imperialism* of high-modernist, planned social order. I stress the word "imperialism" here because I am emphatically not making a blanket case against either bureaucratic planning or high-modernist ideology. I am, however, making a case against an imperial or hegemonic planning mentality that excludes the necessary role of local knowledge and know-how.

Throughout the book I make the case for the indispensable role of practical knowledge, informal processes, and improvisation in the face of unpredictability. In chapters 4 and 5, I contrast the high-modernist views and practices of city planners and revolutionaries with critical views emphasizing process, complexity, and open-endedness. Le Corbusier and Lenin are the protagonists, with Jane Jacobs and Rosa Luxemburg cast as their formidable critics. Chapters 6 and 7 contain accounts of Soviet collectivization and Tanzanian forced villagization, which illustrate how schematic, authoritarian solutions to production and social order inevitably fail when they exclude the fund of valuable knowledge embodied in local practices. (An early draft contained a case study of the Tennessee Valley Authority, the United States' high-modernist experiment and the granddaddy of all regional development projects. It was reluctantly swept aside to shorten what is still a long book.)

Finally, in chapter 9 I attempt to conceptualize the nature of practical knowledge and to contrast it with more formal, deductive, epistemic knowledge. The term *mētis*, which descends from classical Greek and denotes the knowledge that can come only from practical experience, serves as a useful portmanteau word for what I have in mind.

Here I should also acknowledge my debt to anarchist writers (Kropotkin, Bakunin, Malatesta, Proudhon) who consistently emphasize the role of mutuality as opposed to imperative, hierarchical coordination in the creation of social order. Their understanding of the term "mutuality" covers some, but not all, of the same ground that I mean to cover with "mētis."

Radically simplified designs for social organization seem to court the same risks of failure courted by radically simplified designs for natural environments. The failures and vulnerability of monocrop commercial forests and genetically engineered, mechanized monocropping mimic the failures of collective farms and planned cities. At this level, I am making a case for the resilience of both social and natural diversity and a strong case about the limits, in principle, of what we are likely to know about complex, functioning order. One could, I think, successfully turn this argument against a certain kind of reductive social science. Having already taken on more than I could chew, I leave this additional detour to others, with my blessing.

In trying to make a strong, paradigmatic case, I realize that I have risked displaying the hubris of which high modernists are justly accused. Once you have crafted lenses that change your perspective, it is a great temptation to look at everything through the same spectacles. I do, however, want to plead innocent to two charges that I do not think a careful reading would sustain. The first charge is that my argument is uncritically admiring of the local, the traditional, and the customary. I understand that the practical knowledge I describe is often inseparable from the practices of domination, monopoly, and exclusion that offend the modern liberal sensibility. My point is not that practical knowledge is the product of some mythical, egalitarian state of nature. Rather, my point is that formal schemes of order are untenable without some elements of the practical knowledge that they tend to dismiss. The second charge is that my argument is an anarchist case against the state itself. The state, as I make abundantly clear, is the vexed institution that is the ground of both our freedoms and our unfreedoms. My case is that certain kinds of states, driven by utopian plans and an authoritarian disregard for the values, desires, and objections of their subjects, are indeed a mortal threat to human well-being. Short of that draconian but all too common situation, we are left to weigh judiciously the benefits of certain state interventions against their costs.

As I finished this book, I realized that its critique of certain forms of state action might seem, from the post–1989 perspective of capitalist triumphalism, like a kind of quaint archaeology. States with the pretensions and power that I criticize have for the most part vanished or

have drastically curbed their ambitions. And yet, as I make clear in examining scientific farming, industrial agriculture, and capitalist markets in general, large-scale capitalism is just as much an agency of homogenization, uniformity, grids, and heroic simplification as the state is, with the difference being that, for capitalists, simplification must pay. A market necessarily reduces quality to quantity via the price mechanism and promotes standardization; in markets, money talks, not people. Today, global capitalism is perhaps the most powerful force for homogenization, whereas the state may in some instances be the defender of local difference and variety. (In *Enlightenment's Wake*, John Gray makes a similar case for liberalism, which he regards as self-limiting because it rests on cultural and institutional capital that it is bound to undermine.) The "interruption," forced by widespread strikes, of France's structural adjustments to accommodate a common European currency is perhaps a straw in the wind. Put bluntly, my bill of particulars against a certain kind of state is by no means a case for politically unfettered market coordination as urged by Friedrich Hayek and Milton Friedman. As we shall see, the conclusions that can be drawn from the failures of modern projects of social engineering are as applicable to market-driven standardization as they are to bureaucratic homogeneity.

1 Nature and Space

> Would it not be a great satisfaction to the king to know at a designated moment every year the number of his subjects, in total and by region, with all the resources, wealth & poverty of each place; [the number] of his nobility and ecclesiastics of all kinds, of men of the robe, of Catholics and of those of the other religion, all separated according to the place of their residence? . . . [Would it not be] a useful and necessary pleasure for him to be able, in his own office, to review in an hour's time the present and past condition of a great realm of which he is the head, and be able himself to know with certitude in what consists his grandeur, his wealth, and his strengths?
> —Marquis de Vauban, *proposing an annual census to Louis XIV in 1686*

Certain forms of knowledge and control require a narrowing of vision. The great advantage of such tunnel vision is that it brings into sharp focus certain limited aspects of an otherwise far more complex and unwieldy reality. This very simplification, in turn, makes the phenomenon at the center of the field of vision more legible and hence more susceptible to careful measurement and calculation. Combined with similar observations, an overall, aggregate, synoptic view of a selective reality is achieved, making possible a high degree of schematic knowledge, control, and manipulation.

The invention of scientific forestry in late eighteenth-century Prussia and Saxony serves as something of a model of this process.[1] Although the history of scientific forestry is important in its own right, it is used here as a metaphor for the forms of knowledge and manipulation characteristic of powerful institutions with sharply defined interests, of which state bureaucracies and large commercial firms are perhaps the outstanding examples. Once we have seen how simplification, legibility, and manipulation operate in forest management, we can then explore how the modern state applies a similar lens to urban planning, rural settlement, land administration, and agriculture.

The State and Scientific Forestry: A Parable

> I [Gilgamesh] would conquer in the Cedar Forest. . . . I will set my hand to it and will chop down the Cedar.
> —*Epic of Gilgamesh*

The early modern European state, even before the development of scientific forestry, viewed its forests primarily through the fiscal lens of

revenue needs. To be sure, other concerns—such as timber for shipbuilding, state construction, and fuel for the economic security of its subjects—were not entirely absent from official management. These concerns also had heavy implications for state revenue and security.[2] Exaggerating only slightly, one might say that the crown's interest in forests was resolved through its fiscal lens into a single number: the revenue yield of the timber that might be extracted annually.

The best way to appreciate how heroic was this constriction of vision is to notice what fell outside its field of vision. Lurking behind the number indicating revenue yield were not so much forests as commercial wood, representing so many thousands of board feet of saleable timber and so many cords of firewood fetching a certain price. Missing, of course, were all those trees, bushes, and plants holding little or no potential for state revenue. Missing as well were all those parts of trees, even revenue-bearing trees, which might have been useful to the population but whose value could not be converted into fiscal receipts. Here I have in mind foliage and its uses as fodder and thatch; fruits, as food for people and domestic animals; twigs and branches, as bedding, fencing, hop poles, and kindling; bark and roots, for making medicines and for tanning; sap, for making resins; and so forth. Each species of tree—indeed, each part or growth stage of each species—had its unique properties and uses. A fragment of the entry under "elm" in a popular

In state "fiscal forestry," however, the actual tree with its vast number of possible uses was replaced by an abstract tree representing a volume of lumber or firewood. If the princely conception of the forest was still utilitarian, it was surely a utilitarianism confined to the direct needs of the state.

From a naturalist's perspective, nearly everything was missing from

the state's narrow frame of reference. Gone was the vast majority of flora: grasses, flowers, lichens, ferns, mosses, shrubs, and vines. Gone, too, were reptiles, birds, amphibians, and innumerable species of insects. Gone were most species of fauna, except those that interested the crown's gamekeepers.

From an anthropologist's perspective, nearly everything touching on human interaction with the forest was also missing from the state's tunnel vision. The state did pay attention to poaching, which impinged on its claim to revenue in wood or its claim to royal game, but otherwise it typically ignored the vast, complex, and negotiated social uses of the forest for hunting and gathering, pasturage, fishing, charcoal making, trapping, and collecting food and valuable minerals as well as the forest's significance for magic, worship, refuge, and so on.[4]

If the utilitarian state could not see the real, existing forest for the (commercial) trees, if its view of its forests was abstract and partial, it was hardly unique in this respect. Some level of abstraction is necessary for virtually all forms of analysis, and it is not at all surprising that the abstractions of state officials should have reflected the paramount fiscal interests of their employer. The entry under "forest" in Diderot's *Encyclopédie* is almost exclusively concerned with the *utilité publique* of forest products and the taxes, revenues, and profits that they can be made to yield. The forest as a habitat disappears and is replaced by the forest as an economic resource to be managed efficiently and profitably.[5] Here, fiscal and commercial logics coincide; they are both resolutely fixed on the bottom line.

The vocabulary used to organize nature typically betrays the overriding interests of its human users. In fact, utilitarian discourse replaces the term "nature" with the term "natural resources," focusing on those aspects of nature that can be appropriated for human use. A comparable logic extracts from a more generalized natural world those flora or fauna that are of utilitarian value (usually marketable commodities) and, in turn, reclassifies those species that compete with, prey on, or otherwise diminish the yields of the valued species. Thus, plants that are valued become "crops," the species that compete with them are stigmatized as "weeds," and the insects that ingest them are stigmatized as "pests." Thus, trees that are valued become "timber," while species that compete with them become "trash" trees or "underbrush." The same logic applies to fauna. Highly valued animals become "game" or "livestock," while those animals that compete with or prey upon them become "predators" or "varmints."

The kind of abstracting, utilitarian logic that the state, through its officials, applied to the forest is thus not entirely distinctive. What is distinctive about this logic, however, is the narrowness of its field of vision, the degree of elaboration to which it can be subjected, and above

all, as we shall see, the degree to which it allowed the state to impose that logic on the very reality that was observed.[6]

Social Facts, Raw and Cooked

> Society must be remade before it can be the object of quantification. Categories of people and things must be defined, measures must be interchangeable; land and commodities must be conceived as represented by an equivalent in money. There is much of what Weber called rationalization in this, and also a good deal of centralization.
> —Theodore M. Porter, *"Objectivity as Standardization"*

The administrators' forest cannot be the naturalists' forest. Even if the ecological interactions at play in the forest were known, they would constitute a reality so complex and variegated as to defy easy shorthand description. The intellectual filter necessary to reduce the complexity to manageable dimensions was provided by the state's interest in commercial timber and revenue.

If the natural world, however shaped by human use, is too unwieldy in its "raw" form for administrative manipulation, so too are the actual social patterns of human interaction with nature bureaucratically indigestible in their raw form. No administrative system is capable of representing *any* existing social community except through a heroic and greatly schematized process of abstraction and simplification. It is not simply a question of capacity, although, like a forest, a human community is surely far too complicated and variable to easily yield its secrets to bureaucratic formulae. It is also a question of purpose. State agents have no interest—nor should they—in describ-

Forging the Tools of Legibility: Popular Measures, State Measures

Nonstate forms of measurement grew from the logic of local practice. As such, they shared some generic features despite their bewildering variety—features that made them an impediment to administrative uniformity. Thanks to the synthesis of the medievalist Witold Kula, the reasoning that animated local practices of measurement may be set out fairly succinctly.[37]

Most early measures were human in scale. One sees this logic at work in such surviving expressions as a "stone's throw" or "within earshot" for distances and a "cartload," a "basketful," or a "handful" for volume. Given that the size of a cart or basket might vary from place to place and that a stone's throw might not be precisely uniform from person to person, these units of measurement varied geographically and temporally. Even measures that were apparently fixed might be deceptive. The *pinte* in eighteenth-century Paris, for example, was equivalent to .93 liters, whereas in Seine-en-Montagne it was 1.99 liters and in Precy-sous-Thil, an astounding 3.33 liters. The *aune*, a measure of length used for cloth, varied depending on the material (the unit for silk, for instance, was smaller than that for linen), and across France there were at least seventeen different aunes.[38]

Local measures were also relational or "commensurable."[39] Virtually any request for a judgment of measure allows a range of responses depending on the context of the request. In the part of Malaysia with which I am most familiar, if one were to ask "How far is it to the next village?" a likely response would be "Three rice-cookings." The answer assumes that the questioner is interested in how much time it will take to get there, not how many miles away it is. In varied terrain, of course, distance in miles is an utterly unreliable guide to travel time, especially when the traveler is on foot or riding a bicycle. The answer also expresses time not in minutes—until recently, wristwatches were rare—but in units that are locally meaningful. Everyone knows how long it takes to cook the local rice. Thus an Ethiopian response to a query about how much salt is required for a dish might be "Half as much as to cook a chicken." The reply refers back to a standard that everyone is expected to know. Such measurement practices are irreducibly local, inasmuch as regional differences in, say, the type of rice eaten or the preferred way of cooking chicken will give different results.

Many local units of measurement are tied practically to particular activities. Marathi peasants, as Arjun Appadurai notes, express the desired distance between the onion sets they plant in terms of handbreadths. When one is moving along a field row, the hand is, well, the most handy gauge. In similar fashion, a common measure for twine or rope is the distance between the thumb and elbow because this corresponds with how it is wrapped and stored. As with setting onions, the

process of measuring is embedded in the activity itself and requires no separate operation. Such measurements, moreover, are often approximate; they are only as exact as the task at hand requires.[40] Rainfall may be said to be abundant or inadequate if the context of the query implies an interest in a particular crop. And a reply in terms of inches of rainfall, however accurate, would also fail to convey the desired information; it ignores such vital matters as the timing of the rain. For many purposes, an apparently vague measurement may communicate more valuable information than a statistically exact figure. The cultivator who reports that his rice yield from a plot is anywhere between four and seven baskets is conveying more accurate information, when the focus of attention is on the variability of the yield, than if he reported a ten-year statistical average of 5.6 baskets.

There is, then, no single, all-purpose, correct answer to a question implying measurement unless we specify the relevant local concerns that give rise to the question. Particular customs of measurement are thus situationally, temporally, and geographically bound.

Nowhere is the particularity of customary measurement more evident than with cultivated land. Modern abstract measures of land by surface area—so many hectares or acres—are singularly uninformative figures to a family that proposes to make its living from these acres. Telling a farmer only that he is leasing twenty acres of land is about as helpful as telling a scholar that he has bought six kilograms of books. Customary measures of land have therefore taken a variety of forms corresponding to those aspects of the land that are of greatest practical interest. Where land was abundant and manpower or draftpower scarce, the most meaningful gauge of land was often the number of days required to plow or to weed it. A plot of land in nineteenth-century France, for example, would be described as representing so many *morgen* or *journals* (days of work) and as requiring a specific kind of work (*homée, bechée, fauchée*). How many morgen were represented by a field of, say, ten acres could vary greatly; if the land were rocky and steeply pitched, it might require twice as much labor to work than if it were rich bottomland. The morgen would also differ from place to place depending on the strength of local draftpower and the crops sown, and it would differ from time to time as technology (plow tips, yokes, harnesses) affected the work a man could accomplish in a day.

how productive a field had been, although it would reveal little about how arduous the land was to cultivate or how variable the harvests were. But the average yield from a plot of land is itself a rather abstract figure. What most farmers near the subsistence margin want to know above all is whether a particular farm will meet their basic needs reliably. Thus small farms in Ireland were described as a "farm of one cow" or a "farm of two cows" to indicate their grazing capacity to those who lived largely by milk products and potatoes. The physical area a farm might comprise was of little interest compared to whether it would feed a particular family.[41]

To grasp the prodigious variety of customary ways of measuring land, we would have to imagine literally scores of "maps" constructed along very different lines than mere surface area. I have in mind the sorts of maps devised to capture our attention with a kind of fun-house effect in which, say, the size of a country is made proportional to its population rather than its geographical size, with China and India looming menacingly over Russia, Brazil, and the United States, while Libya, Australia, and Greenland virtually disappear. These types of customary maps (for there would be a great many) would construct the landscape according to units of work and yield, type of soil, accessibility, and ability to provide subsistence, none of which would necessarily accord with surface area. The measurements are decidedly *local, interested, contextual,* and *historically specific*. What meets the subsistence needs of one family may not meet the subsistence needs of another. Factors such as local crop regimens, labor supply, agricultural technology, and weather ensure that the standards of evaluation vary from place to place and over time. Directly apprehended by the state, so many maps would represent a hopelessly bewildering welter of local standards. They definitely would not lend themselves to aggregation into a single statistical series that would allow state officials to make meaningful comparisons.

The Politics of Measurement

Thus far, this account of local measurement practices risks giving the impression that, although local conceptions of distance, area, volume, and so on were different from and more varied than the unitary abstract standards a state might favor, they were nevertheless aiming at objective accuracy. That impression would be false. Every act of measurement was an act marked by the play of power relations. To understand measurement practices in early modern Europe, as Kula demonstrates, one must relate them to the contending interests of the major estates: aristocrats, clergy, merchants, artisans, and serfs.

A good part of the politics of measurement sprang from what a contemporary economist might call the "stickiness" of feudal rents. Noble